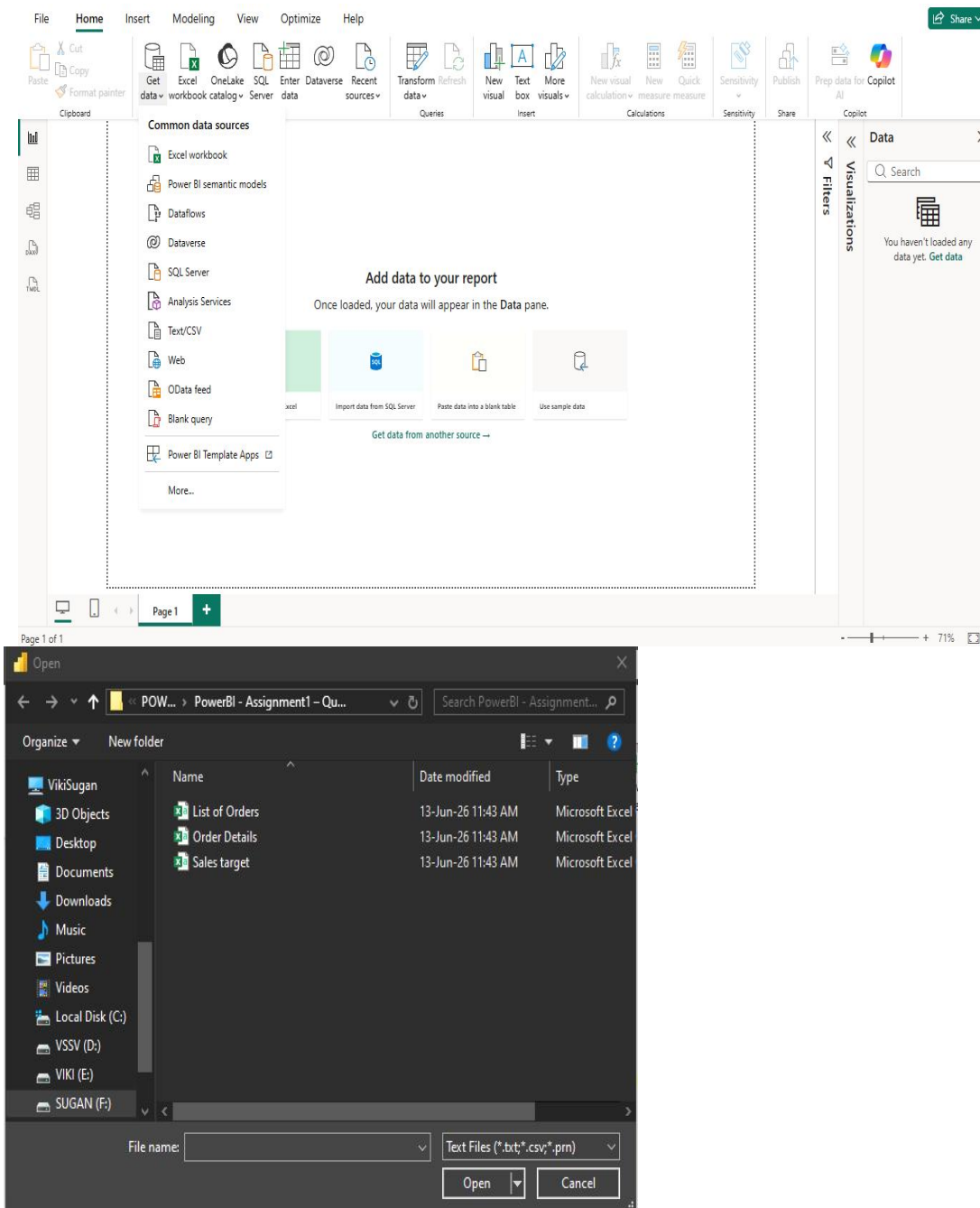


Tasks

Data Import & Transformation

1. Data Import:

- Import *List of Orders.csv* into Power BI.
- Open *the list of orders* in the **Power Query Editor** using the **Transform Data** option.
- Import *Order Details.csv* and *Sales Target.csv* into Power Query Editor.



Microsoft Power Query Editor interface showing a table with columns: Month of Order Date, Category, and Target. The table has 36 rows. The 'Query Settings' pane on the right shows the 'Name' field set to 'Sales target' and the 'Applied Steps' list containing 'Source', 'Promoted Headers', and 'Changed Type'.

Month of Order Date	Category	Target
18-Apr-26	Furniture	10400
18-May-26	Furniture	10500
18-Jun-26	Furniture	10600
18-Jul-26	Furniture	10800
18-Aug-26	Furniture	10900
18-Sep-26	Furniture	11000
18-Oct-26	Furniture	11100
18-Nov-26	Furniture	11300
18-Dec-26	Furniture	11400
19-Jan-26	Furniture	11500
19-Feb-26	Furniture	11600
19-Mar-26	Furniture	11800
18-Apr-26	Clothing	12000
18-May-26	Clothing	12000
18-Jun-26	Clothing	12000
18-Jul-26	Clothing	14000
18-Aug-26	Clothing	14000
18-Sep-26	Clothing	14000
18-Oct-26	Clothing	16000
18-Nov-26	Clothing	16000
18-Dec-26	Clothing	16000
19-Jan-26	Clothing	16000

2. Row Limiting & Data Types:

- Restrict the *List of Orders* table to the **first 500 rows**.
- Set the **Order Date** column to the **Date** data type.
- Change the **Amount** and **Target** columns to **fixed decimal numbers**.

File Home Transform Add Column View Tools Help

Group By Use First Row as Headers Count Rows

Table

Data Type: Text Replace Values Unpivot Columns

Detect Data Type Fill Move

Rename Pivot Column Convert to List

Any Column

Split Column Format Merge Columns

Text Column

Statistics Standard Scientific

Number Column

Queries [3]

List of Orders

Order Details

Sales target

fx = Table.TransformColumnTypes(Source,{{"Column1", type text}, {"Column2", type text}, {"Column3", type text},

	Column1	Column2	Column3	Column4	Column5
	Valid 89%	Valid 89%	Valid 89%	Valid 89%	Valid 89%
	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%
	Empty 11%	Empty 11%	Empty 11%	Empty 11%	Empty 11%
1	Order ID	Order Date	CustomerName	State	City
2	B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad
3	B-25602	01-04-2018	Pearl	Maharashtra	Pune

File Home Transform Add Column View Tools Help

Group By Use First Row as Headers Count Rows

Table

Data Type: Text Replace Values Unpivot Columns

Detect Data Type Fill Move

Rename Pivot Column Convert to List

Any Column

Split Column Format Merge Columns

Text Column

Statistics Standard Scientific

Number Column

Queries [3]

List of Orders

Order Details

Sales target

fx = Table.PromoteHeaders("#Changed Type", [PromoteAllScalars=true])

	Order ID	Order Date	CustomerName	State	City
	Valid 89%	Valid 89%	Valid 89%	Valid 89%	Valid 89%
	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%
	Empty 11%	Empty 11%	Empty 11%	Empty 11%	Empty 11%
1	B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad
2	B-25602	01-04-2018	Pearl	Maharashtra	Pune

fx = Table.TransformColumns("#Cleaned Text-Order ID",{{"Order ID", Text.Trim, type text}})

	Order ID	Order Date	CustomerName	State	City
	Valid 89%	Valid 89%	Valid 89%	Valid 89%	Valid 89%
	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%
	Empty 11%	Empty 11%	Empty 11%	Empty 11%	Empty 11%
1	B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad
2	B-25602	01-04-2018	Pearl	Maharashtra	Pune

Keep Range of Rows

Specify the range of rows to keep.

First row

1

Number of rows

500

OK

Enter Data | Data source settings | Manage Parameters | Export query results | Refresh Preview | Properties | Advanced Editor | Choose Columns | Remove Columns | Keep Rows | Remove Rows | Split Column | Group By | Data Type | Use | Rep | Transf

Query

Manage Columns

Keep Top Rows
Keep Bottom Rows
Keep Range of Rows
Keep Duplicates
Keep Errors

Table.PromoteHeaders("#Changed Type", [PromoteAllScalars=true])

Order ID	Order Date	CustomerName	State
497 B-26097	28-03-2019	Vini	Karnataka
498 B-26098	29-03-2019	Pinky	Jammu and Kashmir
499 B-26099	30-03-2019	Bhishm	Maharashtra
500 B-26100	31-03-2019	Hitika	Madhya Pradesh

Data Type: Text | Replace Values | Unpivot Columns | Move | Convert to List

Decimal Number
Fixed decimal number
Whole Number
Percentage
Date/Time
Date
Time
Date/Time/Timezone
Duration
Text
True/False
Binary

Amount	Profit
1,275.00	
66.00	
8.00	
80.00	
168.00	
424.00	

\$ Amount	
Valid	100%
Error	0%
Empty	0%
	1,275.00
	66.00
	8.00
	80.00
	168.00
	424.00

Quantity | Category | Sub-Category

Copy
Remove
Remove Other Columns
Duplicate Column
Add Column From Examples...
Remove Duplicates
Remove Errors
Change Type
Transform
Replace Values...
Replace Errors...
Split Column
Group By...
Fill
Unpivot Columns
Unpivot Other Columns
Unpivot Only Selected Columns
Rename...
Move
Drill Down
Add as New Query

PROPERTIES
Name
Order Details
All Properties
APPLIED STEPS
Source
Promoted Headers
Changed Type

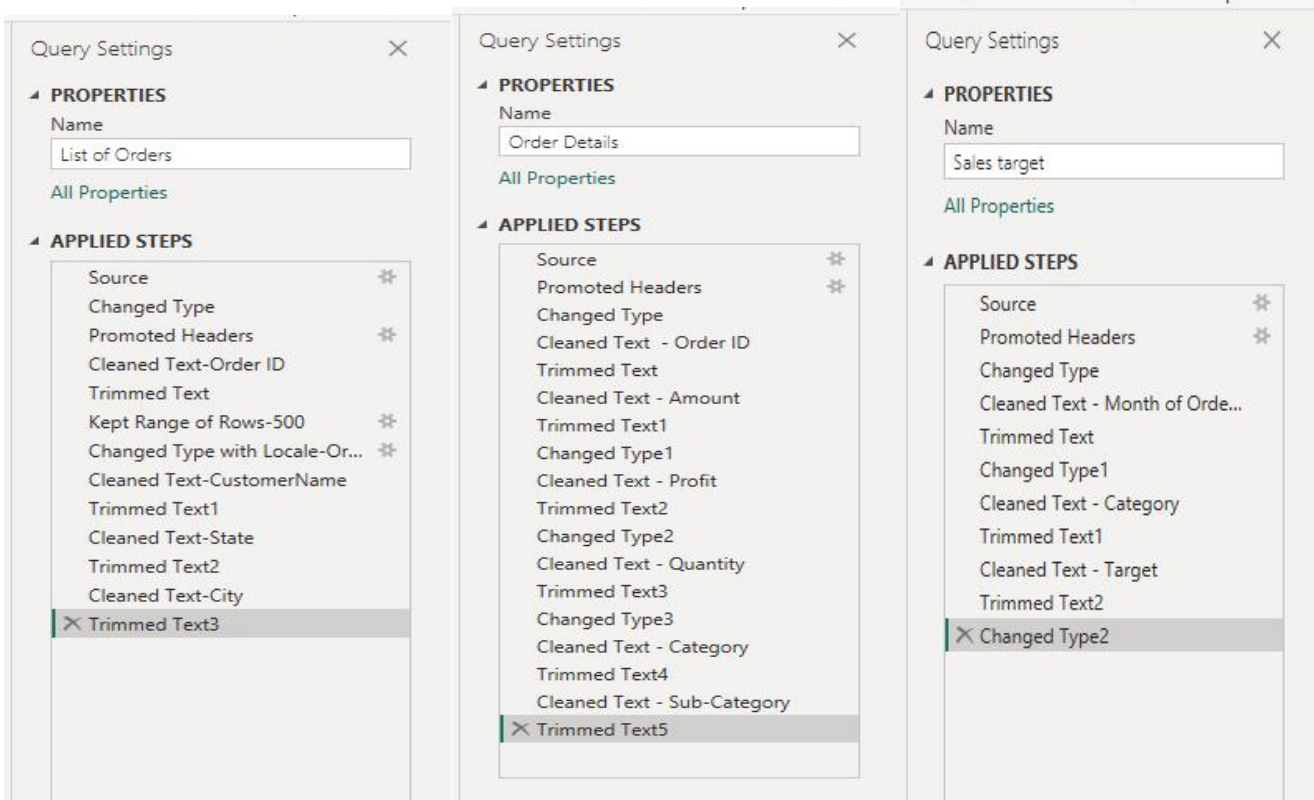
Decimal Number
Fixed decimal number
Whole Number
Percentage
Date/Time
Date
Time
Date/Time/Timezone
Duration
Text
True/False
Binary
Using Locale...

Data Type: Fixed decimal number | Replace Values | Unpivot Columns | Fill | Move | Convert to List | Split Column | Format | Merge | Parse

Detect Data Type
Rename
Any Column

Table.TransformColumnTypes("#Trimmed Text2",{{"Target", Currency.Type}})

Month of Order Date	Category	Target
18-Apr-26	Furniture	10,400.00
18-May-26	Furniture	10,500.00
18-Jun-26	Furniture	10,600.00



3. Text Formatting:

- Format the **CustomerName** column to **proper case** to ensure consistent capitalization.

	A ^B C	CustomerName	A ^B C
100%	Valid	100%	
0%	Error	0%	
0%	Empty	0%	
~18	Bharat	Guj	
~18	Pearl	Mal	
~18	Jahan	Ma	
~18	Divsha	Raj	
~18	Kasheen	We	
~18	Hazel	Kar	
~18	Sonakshi	Jam	
~18	Aarushi	Tan	
~18	Jitesh	Utt	
~18	Yogesh	Bihi	
~18	Anita	Ker	

Query Settings

PROPERTIES

Name: List of Orders

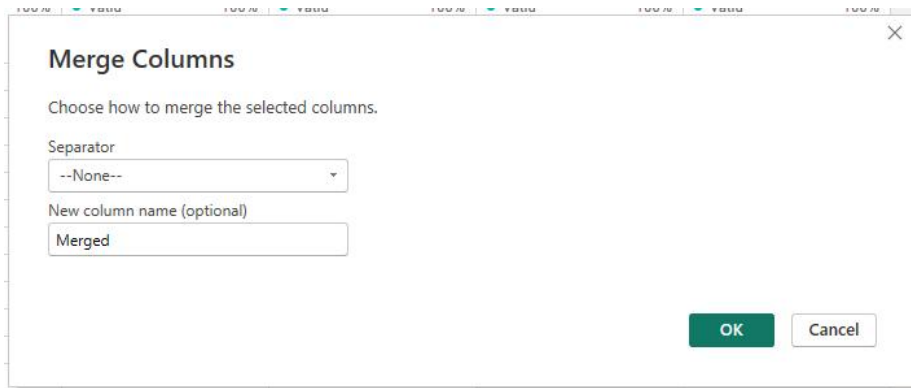
All Properties

APPLIED STEPS

- Source
- Changed Type
- Promoted Headers
- Cleaned Text-Order ID
- Trimmed Text
- Kept Range of Rows-500
- Changed Type with Locale-Or...
- Cleaned Text - CustomerName
- Trimmed Text1
- Capitalized Each Word
- Cleaned Text - State
- Trimmed Text2
- Changed Type2
- Cleaned Text - City
- Trimmed Text3
- Changed Type1**

4. Column Creation:

- Merge the **City** and **State** columns to create a new column named **Location** in the format: *City, State*



A dialog box titled "Merge Columns" with a close button (X) in the top right corner. The text inside says "Choose how to merge the selected columns." Below this, there is a "Separator" dropdown menu currently set to "--None--". Underneath is a text input field for "New column name (optional)" containing the word "Merged". At the bottom right are "OK" and "Cancel" buttons.

```
,{{"Location", Text.Trim, type text}}}
```

	A ^B C State		A ^B C City		A ^B C Location	
100%	Valid	100%	Valid	100%	Valid	100%
0%	Error	0%	Error	0%	Error	0%
0%	Empty	0%	Empty	0%	Empty	0%
	Gujarat		Ahmedabad		Ahmedabad,Gujarat	
	Maharashtra		Pune		Pune,Maharashtra	
	Madhya Pradesh		Bhopal		Bhopal,Madhya Pradesh	
	Rajasthan		Jaipur		Jaipur,Rajasthan	
	West Bengal		Kolkata		Kolkata,West Bengal	
	Karnataka		Bangalore		Bangalore,Karnataka	
	Jammu and Kashmir		Kashmir		Kashmir,Jammu and Kashmir	
	Tamil Nadu		Chennai		Chennai,Tamil Nadu	
	Uttar Pradesh		Lucknow		Lucknow,Uttar Pradesh	
	Other		Other		Other,Other	

- Create a custom column named **Profit Margin** calculated as: *Profit / Amount* (expressed as a percentage)

File Home Transform Add Column View

Column From Examples Custom Column Invoke Custom Function Conditional Column Index Column Duplicate Column

General

Queries [3]

- List of Orders
- Order Details
- Sales target

Order ID

- Valid
- Error
- Empty

1	B-25601
2	B-25601

Custom Column

Add a column that is computed from the other columns.

New column name
Profit Margin

Custom column formula
= [Profit]/[Amount]

Available columns

- Order ID
- Amount
- Profit
- Profit Margin
- Quantity
- Category
- Sub-Category

<< Insert

Learn about Power Query formulas

✓ No syntax errors have been detected.

OK Cancel

Transform Add Column View Tools Help

Transpose Reverse Rows Count Rows

Data Type: Percentage Replace Values

- Decimal Number
- Fixed decimal number
- Whole Number
- Percentage
- Date/Time
- Date
- Time
- Date/Time/Timezone
- Duration
- Text
- True/False
- Binary

1	B-25602
2	B-25602
3	B-25602
4	B-25602
5	B-25602
6	B-25602
7	B-25602

\$ Amount	\$ Profit	% Profit Margin	123 Qu
Valid 100%	Valid 100%	Valid 100%	Valid
Error 0%	Error 0%	Error 0%	Error
Empty 0%	Empty 0%	Empty 0%	Empty
1,275.00	-1,148.00	-90.04%	
66.00	-12.00	-18.18%	
8.00	-2.00	-25.00%	
80.00	-56.00	-70.00%	
168.00	-111.00	-66.07%	
424.00	-272.00	-64.15%	
2,617.00	1,151.00	43.98%	
561.00	212.00	37.79%	
119.00	-5.00	-4.20%	
1,355.00	-60.00	-4.43%	

5. Conditional Column:

○ Add a conditional column named **Profit Status** based on:

- Profit < 0 → *Loss*
- Profit = 0 → *Break-Even*
- Profit > 0 → *Profit*

Add Conditional Column

Add a conditional column that is computed from the other columns or values.

New column name: Profit Status

Column Name	Operator	Value	Output
If Profit	is less than	0	Loss
Else If Profit	equals	0	Break - Even
Else If Profit	is greater than	0	Profit

Buttons: Add Clause, OK, Cancel

Transform | Add Column | View | Tools | Help

Formula: `= Table.TransformColumnTypes("#Reordered Columns1",{{"Profit Status", type text}})`

	Amount	Profit	Profit Margin	Profit Status
1	1,275.00	-1,148.00	-90.04%	Loss
2	66.00	-12.00	-18.18%	Loss
3	8.00	-2.00	-25.00%	Loss
4	80.00	-56.00	-70.00%	Loss
5	168.00	-111.00	-66.07%	Loss
6	424.00	-272.00	-64.15%	Loss
7	2,617.00	1,151.00	43.98%	Profit
8	561.00	212.00	37.79%	Profit
9	119.00	-5.00	-4.20%	Loss
10	1,355.00	-60.00	-4.43%	Loss

6. Merging Data (Joins)

- Merge the *List of Orders* and *Order Details* tables using **Order ID**.
- Name the merged output table as **Orders Data**.

Merge

Select tables and matching columns to create a merged table.

List of Orders

Order ID	Order Date	CustomerName	State	City	Location
B-25601	01-Apr-18	Bharat	Gujarat	Ahmedabad	Ahmedabad,Gujarat
B-25602	01-Apr-18	Pearl	Maharashtra	Pune	Pune,Maharashtra
B-25603	03-Apr-18	Jahan	Madhya Pradesh	Bhopal	Bhopal,Madhya Pradesh
B-25604	03-Apr-18	Divsha	Rajasthan	Jaipur	Jaipur,Rajasthan
B-25605	05-Apr-18	Kasheen	West Bengal	Kolkata	Kolkata,West Bengal

Order Details

Order ID	Amount	Profit	Profit Margin	Profit Status	Quantity	Category	Sub-Category
B-25601	1,275.00	-1,148.00	-90.04%	Loss	7	Furniture	Bookcases
B-25601	66.00	-12.00	-18.18%	Loss	5	Clothing	Stole
B-25601	8.00	-2.00	-25.00%	Loss	3	Clothing	Hankerchief
B-25601	80.00	-56.00	-70.00%	Loss	4	Electronics	Electronic Games
B-25602	168.00	-111.00	-66.07%	Loss	2	Electronics	Phones

Join Kind

Left Outer (all from first, matching from second)

☐ Use fuzzy matching to perform the merge

► Fuzzy matching options

✓ The selection matches 500 of 500 rows from the first table.

OK Cancel

FileHomeTransformAdd ColumnViewToolsHelp

Close & Apply

New Source

Recent Sources

Enter Data

Data source settings

Manage Parameters

Export query results

Refresh Preview

Properties

Advanced Editor

Manage

Choose Columns

Remove Columns

Keep Rows

Remove Rows

Sort

Split Column

Group By

Replace Values

Data Type: Text

Use First Row as Header

Close

New Query

Data Sources

Parameters

Output Data

Query

Manage Columns

Reduce Rows

Transform

Queries [4]

×

✓

f_x

= Table.NestedJoin("#List of Orders", {"Order ID"}, #Order Details", {"Order ID"}, "Order Details", JoinKind.LeftOuter)

List of Orders

Order Details

Sales target

Order Data

	A ^B _C Order ID	Order Date	A ^B _C CustomerName	A ^B _C State	A ^B _C City	A ^B _C Location
	Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid 100%
	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%
	Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty 0%
1	B-25601	01-Apr-18	Bharat	Gujarat	Ahmedabad	Ahmedabad,Gujarat
2	B-25602	01-Apr-18	Pearl	Maharashtra	Pune	Pune,Maharashtra
3	B-25603	03-Apr-18	Jahan	Madhya Pradesh	Bhopal	Bhopal,Madhya Pradesh
4	B-25604	03-Apr-18	Divsha	Rajasthan	Jaipur	Jaipur,Rajasthan
5	B-25605	05-Apr-18	Kasheen	West Bengal	Kolkata	Kolkata,West Bengal
6	B-25606	06-Apr-18	Hazel	Karnataka	Bangalore	Bangalore Karnataka

7. Handling Missing Data & Duplicate Data

- Identify missing values across all tables.
- Define and apply appropriate strategies such as:
 - Replacing missing values
 - Removing records
 - Retaining nulls with justification
- Identify duplicate rows and define a suitable handling strategy based on business logic.

For Handling Missing Data & Duplicate Data in Power BI (Power Query Editor), you can document and perform the cleaning process like this:

Handling Missing Data & Duplicate Data

1. Identify Missing Values Across All Tables

Steps:

Open Power BI Desktop

Go to Home → Transform Data

Select each table one by one

Review columns for:

null

blank values

empty cells

You can also:

Go to View → Column Quality

Enable Column Distribution and Column Profile

These help identify missing values quickly.

2. Apply Missing Value Handling Strategy

A. Replace Missing Values

Use when values can reasonably be defaulted.

Example:

Discount Column

Replace null $\rightarrow$ 0

Steps:

Select column

Transform $\rightarrow$ Replace Values

Replace:

null $\rightarrow$ 0

Example:

Quantity Column

Replace null $\rightarrow$ 1

Reason:

Assume minimum order quantity.

Steps:

Select column

Transform $\rightarrow$ Replace Values

null $\rightarrow$ 1

B. Remove Records

Use when critical information is missing.

Example:

If Order ID is missing:

Steps:

Select Order ID

Home → Remove Rows → Remove Blank Rows

Reason:

Records without unique identifiers cannot be matched or analyzed.

C. Retain Null Values (with justification)

Use when missing data carries meaning.

Example:

Customer Feedback

Keep null values

Reason:

Blank may indicate customer did not provide feedback.

No action required.

3. Identify Duplicate Rows

Steps:

Select table

Select all columns or key columns

Go to:

Home → Remove Rows → Remove Duplicates

Suggested Documentation Statement

Missing values were identified using Column Quality and handled using replacement, removal, or retention strategies depending on business relevance. Duplicate rows were evaluated using business rules and removed only when they represented unintended repeated records.

8. Sorting and Filtering Data

- Sort records in the **Orders Data** table by **Order Date** in descending order.

File

Tools

Help

Manage Parameters

Export query results

Refresh Preview

Properties

Advanced Editor

Manage

Choose Columns

Remove Columns

Keep Rows

Remove Rows

Sort

Query

Manage Columns

Reduce Rows

Table

Order Date

CustomerName

State

City

0% Valid 100%

0% Error 0%

0% Empty 0%

01-Apr-18

Bharat

Gujarat

Ahmedabad

01-Apr-18

Pearl

Maharashtra

Pune

03-Apr-18

Jahan

Madhya Pradesh

Bhopal

03-Apr-18

Divsha

Rajasthan

Jaipur

05-Apr-18

Kasheen

West Bengal

Kolkata

06-Apr-18

Hazel

Karnataka

Bangalore

06-Apr-18

Sonakshi

Jammu and Kashmir

Kashmir

08-Apr-18

Arushi

Tamil Nadu

Chennai

Table

Order Date

CustomerName

100% Valid 100%

0% Error 0%

0% Empty 0%

31-Mar-19

Hitika

30-Mar-19

Bhishm

29-Mar-19

Pinky

28-Mar-19

Vini

28-Mar-19

Atharv

28-Mar-19

Monisha

27-Mar-19

Deepak

27-Mar-19

Manju

27-Mar-19

Ramesh

27-Mar-19

Sarita

27-Mar-19

Sagar

26-Mar-19

Bhavna

26-Mar-19

Mukesh

26-Mar-19

Vandana

26-Mar-19

Shrichand

26-Mar-19

Kanak

25-Mar-19

Anita

24-Mar-19

Yogesh

23-Mar-19

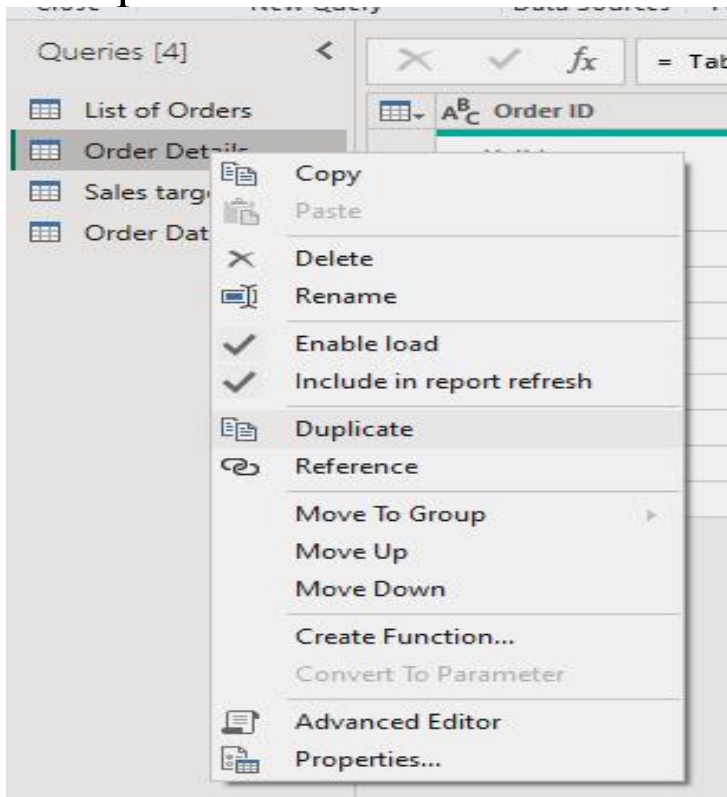
Jitesh

- Apply filters to analyze orders from a specific state (e.g., **Tamil Nadu**).

	A ^B _C Order ID	Order Date	A ^B _C CustomerName	A ^B _C State	A ^B _C City	A ^B _C Location
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
1	B-26081	22-Mar-19	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu
2	B-26018	14-Feb-19	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu
3	B-26008	09-Feb-19	Kalyani	Tamil Nadu	Chennai	Chennai,Tamil Nadu
4	B-25860	15-Nov-18	Akshay	Tamil Nadu	Chennai	Chennai,Tamil Nadu
5	B-25788	21-Sep-18	Dinesh	Tamil Nadu	Chennai	Chennai,Tamil Nadu
6	B-25716	11-Jul-18	Surabhi	Tamil Nadu	Chennai	Chennai,Tamil Nadu
7	B-25698	23-Jun-18	Amisha	Tamil Nadu	Chennai	Chennai,Tamil Nadu
8	B-25608	08-Apr-18	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu

9. Grouping and Aggregating Data

- Duplicate the *Order Details* table and perform the following:



○ Count of Order ID

Group By

Specify the column to group by and the desired output.

☒ Basic ☐ Advanced

Order Count

New column name: Order Count Operation: Count Rows Column:

OK Cancel

= Table.Group("#Changed Type5", {"Order ID"			
A ^B	Order ID	1 ²	Order Count
Valid	100%	Valid	100%
Error	0%	Error	0%
Empty	0%	Empty	0%
1	B-25601		4
2	B-25602		5
3	B-25603		8
4	B-25604		2
5	B-25605		1
6	B-25606		1
7	B-25607		1
8	B-25608		4
9	B-25609		2
10	B-25610		6
11	B-25611		1
12	B-25612		1
13	B-25613		1
14	B-25614		2
15	B-25615		1
16	B-25616		4
17	B-25617		1
18	B-25618		2

○ Average Profit by Category

Group By

Specify the column to group by and the desired output.

☒ Basic ☐ Advanced

Category

New column name: Average Profit Operation: Average Column: Profit

OK Cancel

Queries [6] <

✕ ✓ *fx* = Table.Group(#"Changed Type5", {"Catego

	A ^B C Category	1.2 Average Profit
	- Valid 100% - Error 0% - Empty 0%	- Valid 100% - Error 0% - Empty 0%
1	Furniture	9.456790123
2	Clothing	11.76290832
3	Electronics	34.07142857

○ Total Amount by Sub-Category

Group By

✕

Specify the column to group by and the desired output.

☒ Basic
 ☐ Advanced

Sub-Category

▼

New column name

Operation

Column

Total Amount

Sum

Amount

OK

Cancel

Queries [7] <

✕ ✓ *fx* = Table.Group(#"Changed Type5", {"Sub-Ca

	A ^B _C Sub-Category	1.2 Total Amount
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
1	Bookcases	56861
2	Stole	18546
3	Hankerchief	14608
4	Electronic Games	39168
5	Phones	46119
6	Saree	53511
7	Trousers	30039
8	Chairs	34222
9	Kurti	3361
10	T-shirt	7382
11	Shirt	7555
12	Leggings	2106
13	Tables	22614
14	Printers	58252
15	Accessories	21728
16	Furnishings	13484
17	Skirt	1946

- Duplicate the *Sales Target* table and aggregate the total Target by **Month of order date**.

FileHomeTransformAdd ColumnViewToolsHelp

Column From Examples

Custom Column

Invoke Custom Function

Conditional Column

Index Column

Duplicate Column

Format

From Text

Merge Columns

Extract

Parse

Statistics

Standard

Scientific

10²

Rounding

Information

Trigonometry

Date

Time

Duration

Queries [8]

fx

= Table.TransformColumnTypes(#"Trimmed Text2",{{"Target", Currency.Type}})

Month of Order Date

ABC Category

\$ Target

Valid

Error

Empty

100%

0%

0%

Valid

Error

Empty

100%

0%

0%

Valid

Error

Empty

100%

0%

0%

1	18-Apr-26	Furniture	10,400.00
2	18-May-26	Furniture	10,500.00
3	18-Jun-26	Furniture	10,600.00
4	18-Jul-26	Furniture	10,800.00
5	18-Aug-26	Furniture	10,900.00
6	18-Sep-26	Furniture	11,000.00
7	18-Oct-26	Furniture	11,100.00
8	18-Nov-26	Furniture	11,300.00
9	18-Dec-26	Furniture	11,400.00

Age

Date Only

Parse

Year

Month

Quarter

Week

Day

Subtract Days

Combine Date and Time

Earliest

Latest

Month

Start of Month

End of Month

Days in Month

Name of Month

Queries [8]

fx

= Table.AddColumn(#"Changed Type2", "Month Name", each Date.MonthName([Month of Order Date]))

Month of Order Date

ABC Category

\$ Target

ABC Month Name

Valid

Error

Empty

100%

0%

0%

Valid

Error

Empty

100%

0%

0%

Valid

Error

Empty

100%

0%

0%

Valid

Error

Empty

100%

0%

0%

1	18-Apr-26	Furniture	10,400.00	April
2	18-May-26	Furniture	10,500.00	May
3	18-Jun-26	Furniture	10,600.00	June
4	18-Jul-26	Furniture	10,800.00	July
5	18-Aug-26	Furniture	10,900.00	August
6	18-Sep-26	Furniture	11,000.00	September
7	18-Oct-26	Furniture	11,100.00	October
8	18-Nov-26	Furniture	11,300.00	November
9	18-Dec-26	Furniture	11,400.00	December
10	19-Jan-26	Furniture	11,500.00	January
11	19-Feb-26	Furniture	11,600.00	February
12	19-Mar-26	Furniture	11,800.00	March
13	18-Apr-26	Clothing	12,000.00	April
14	18-May-26	Clothing	12,000.00	May
15	18-Jun-26	Clothing	12,000.00	June
16	18-Jul-26	Clothing	14,000.00	July
17	18-Aug-26	Clothing	14,000.00	August
18	18-Sep-26	Clothing	14,000.00	September
19	18-Oct-26	Clothing	16,000.00	October
20	18-Nov-26	Clothing	16,000.00	November
21	18-Dec-26	Clothing	16,000.00	December
22	19-Jan-26	Clothing	16,000.00	January

4 COLUMNS, 36 ROWS Column profiling based on top 1000 rows



Group By

Specify the column to group by and the desired output.

☒ Basic ☐ Advanced

Month

New column name

Total Target

Operation

Sum

Column

Target

OK

Cancel

Queries [8]



= Table.Group("#Renamed Columns", {"Month

- List of Orders
- Order Details
- Sales target
- Order Data
- Order Details Sum...
- Order - Profit
- Total Amount
- Sales target - Sum...

	A ^B C Month		1.2 Total Target	
	Valid	100%	Valid	100%
	Error	0%	Error	0%
	Empty	0%	Empty	0%
1	April		31400	
2	May		31500	
3	June		31600	
4	July		33800	
5	August		33900	
6	September		34000	
7	October		36100	
8	November		36300	
9	December		36400	
10	January		43500	
11	February		43600	
12	March		43800	

10. Data Modeling:

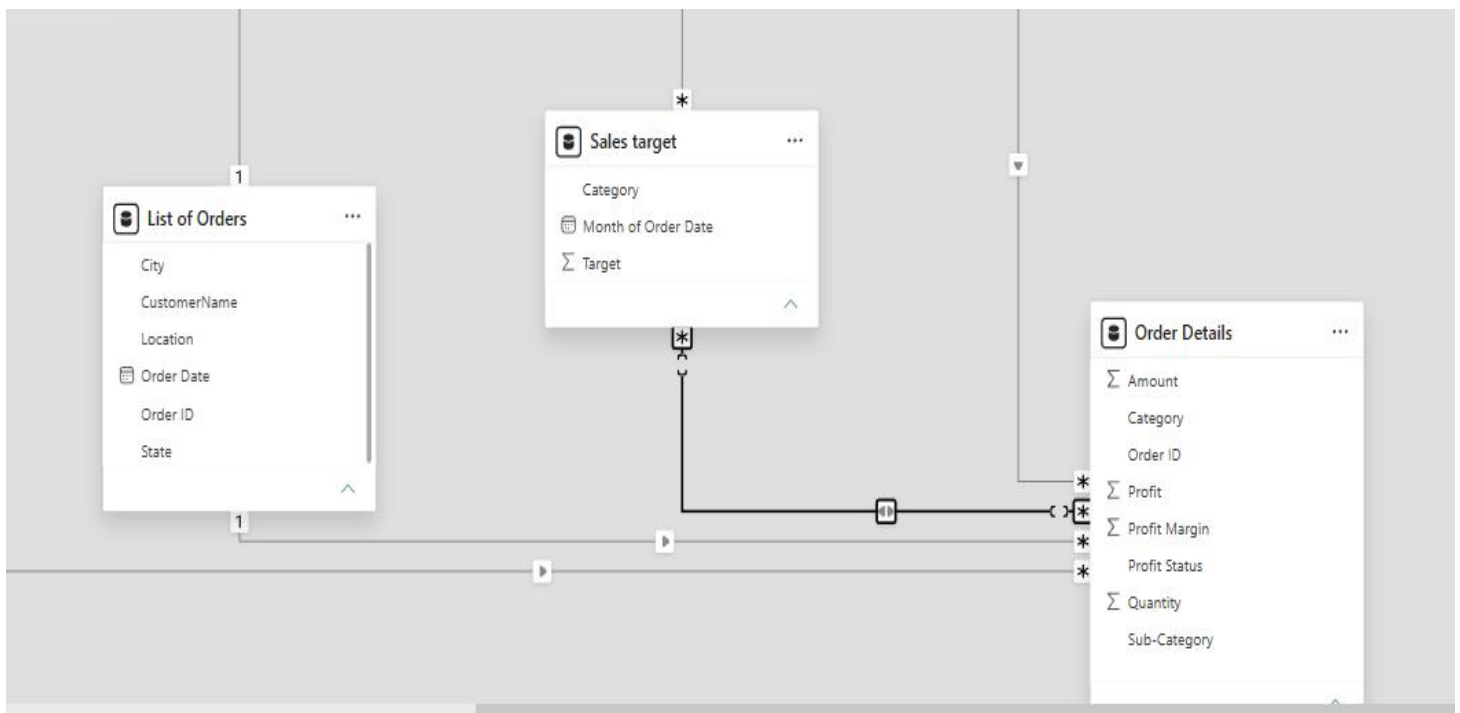
- Establish a relationship between:
 - *List of Orders* and *Order Details* using **Order ID**

☐	From: table (column) ↑	Relationship	To: table (column)	Status
☐	Order Details (Category)		Order - Profit (Category)	Active ...
☐	Order Details (Order ID)		List of Orders (Order ID)	Active ...
☐	Order Details (Sub-Category)		Total Amount (Sub-Category)	Active ...
☐	Order Details - Summary (Ord...		List of Orders (Order ID)	Active ...
☐	Sales target (Category)		Order - Profit (Category)	Active ...

- Establish a relationship between:
 - *Order Details* and *Sales Target* using **Category**

Manage relationships				
+ New relationship Autodetect		Edit Delete Filter		
☐ From: table (column) ↑	Relationship	To: table (column)	Status	
☐ Order Details (Category)	* ← 1	Order - Profit (Category)	Active	...
☒ Order Details (Category)	* ↔ *	Sales target (Category)	Active	...
☒ Order Details (Order ID)	* ← 1	List of Orders (Order ID)	Active	...
☐ Order Details (Sub-Category)	* ← 1	Total Amount (Sub-Category)	Active	...
☐ Order Details - Summary (Ord...	1 ↔ 1	List of Orders (Order ID)	Active	...
☐ Sales target (Category)	* ← 1	Order - Profit (Category)	Active	...

2. Use **Manage Relationships** to ensure relationships are active and correctly defined.



DAX Calculations

Calculated Columns

Name	Revenue per Order	\$% Format	Currency	Summarization	Sum	Sort by column
Data type	Fixed decimal num...	\$ %	Auto	Data category	Uncategorized	Sort
Structure	Formatting	Properties				
1	Revenue per Order =					
2	'Order Details'[Amount] * 'Order Details'[Quantity]					

Order ID	Amount	Profit	Quantity	Category	Sub-Category	Category Type	Revenue per Order
B-25602	561	212	3	Clothing	Saree	Clothing - Saree	\$1,683
B-25602	119	-5	8	Clothing	Saree	Clothing - Saree	\$952
B-25603	193	-166	3	Clothing	Saree	Clothing - Saree	\$579
B-25604	157	5	9	Clothing	Saree	Clothing - Saree	\$1,413
B-25605	75	0	7	Clothing	Saree	Clothing - Saree	\$525
B-25609	25	-5	4	Clothing	Saree	Clothing - Saree	\$100
B-25610	43	0	3	Clothing	Saree	Clothing - Saree	\$129
B-25611	160	-59	2	Clothing	Saree	Clothing - Saree	\$320
B-25613	1603	0	9	Clothing	Saree	Clothing - Saree	\$14,427
B-25619	353	90	8	Clothing	Saree	Clothing - Saree	\$2,824
B-25622	534	0	3	Clothing	Saree	Clothing - Saree	\$1,602
B-25623	149	-87	4	Clothing	Saree	Clothing - Saree	\$596
B-25625	635	-349	5	Clothing	Saree	Clothing - Saree	\$3,175

• Create a Calculated Column to Categorize Sales:

Add a calculated column named 'Sales Category' in the Order Details table that categorizes each order as 'Above Average' or 'Below Average' based on the Amount value.

Formula :

Sales Category =

IF(
 'Order Details'[Amount] >
 AVERAGE('Order Details'[Amount]),
 "Above Average",
 "Below Average"
)

1	Sales Category =					
2	IF(					
3	'Order Details'[Amount] >					
4	AVERAGE('Order Details'[Amount]),					
5	"Above Average",					
6	"Below Average"					
7	)					

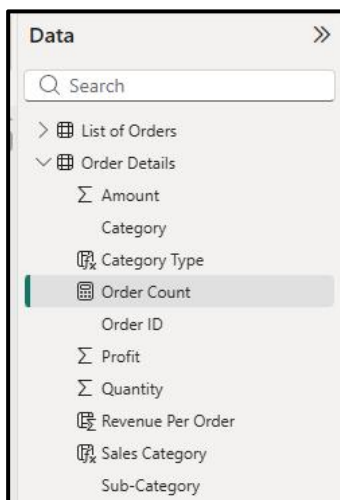
Order ID	Amount	Profit	Quantity	Category	Sub-Category	Category Type	Revenue per Order	Sales Category
B-25602	561	212	3	Clothing	Saree	Clothing - Saree	\$1,683	Above Average
B-25602	119	-5	8	Clothing	Saree	Clothing - Saree	\$952	Below Average
B-25603	193	-166	3	Clothing	Saree	Clothing - Saree	\$579	Below Average
B-25604	157	5	9	Clothing	Saree	Clothing - Saree	\$1,413	Below Average
B-25605	75	0	7	Clothing	Saree	Clothing - Saree	\$525	Below Average
B-25609	25	-5	4	Clothing	Saree	Clothing - Saree	\$100	Below Average
B-25610	43	0	3	Clothing	Saree	Clothing - Saree	\$129	Below Average
B-25611	160	-59	2	Clothing	Saree	Clothing - Saree	\$320	Below Average
B-25613	1603	0	9	Clothing	Saree	Clothing - Saree	\$14,427	Above Average

Calculated Measures

1. Order Count- Create a measure that calculates the **total number of orders** in the **Order Details** table.

Formula :

Order Count = `COUNT('Order Details'[Order ID])`

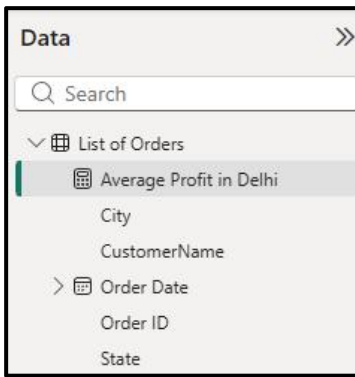


2. Average Profit in Delhi- Create a measure that calculates the **average profit** for orders placed in **Delhi**.

Formula:

Average Profit in Delhi =

```
CALCULATE(  
    AVERAGE('Order Details'[Profit]),  
    'List of Orders'[City] = "Delhi"  
)
```

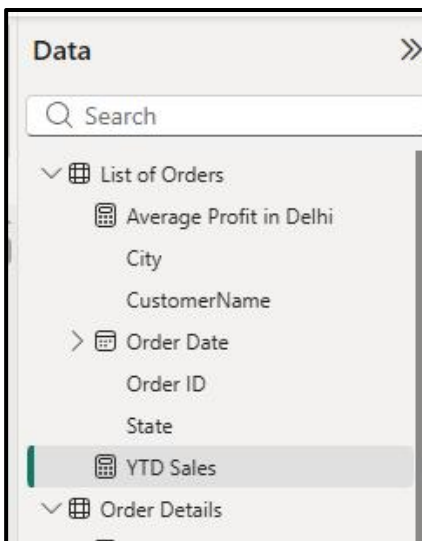


3. **Year-to-Date (YTD) Sales-** Create a measure that calculates **total sales accumulated from the beginning of the year up to the current order date.**

Formula:

YTD Sales =

TOTALYTD(
 SUM('Order Details'[Amount]),
 'List of Orders'[Order Date])



Data Visualization

Create the following visualizations using Power BI:

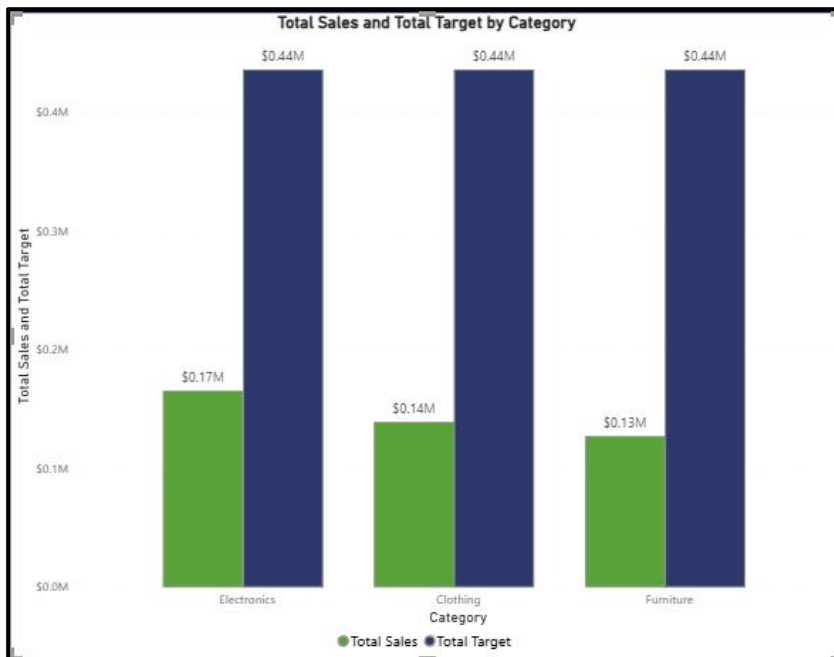
- **Sales Target Achievement by Category:**

Compare actual sales with sales targets by category using a clustered column chart.

Visual: Clustered Column Chart

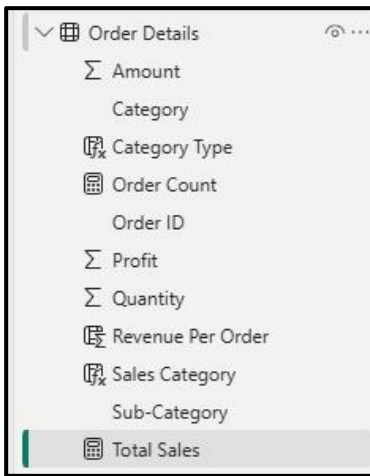
Fields

- **Axis:** Category
- **Values:** Total Sales
- **Values:** Sales Target

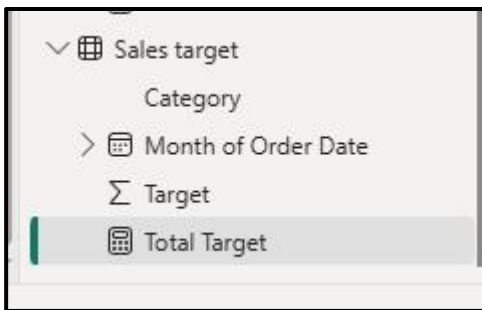


Measures:

Total Sales =SUM('Order Details'[Amount])



Total Target =
SUM('Sales Target'[Target])



Purpose:

- Compare actual vs target.
- Identify categories exceeding targets.

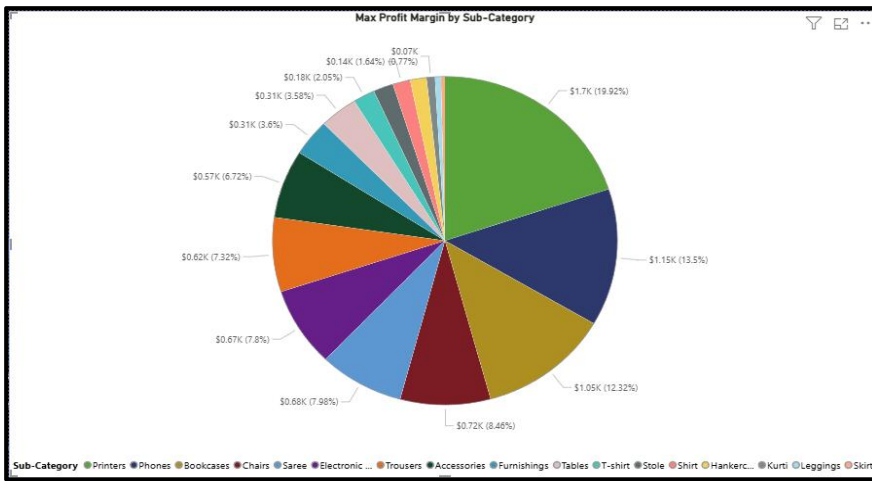
● Max Profit Margin by Sub-Category:

Analyze the maximum profit margin for each sub-category of products using a donut chart.

Visual: Donut Chart

Fields

- Legend → Sub-Category
- Values → Max Profit Margin



Create Measure:

Profit-Margin =

DIVIDE(
 SUM('Order Details'[Profit]),
 SUM('Order Details'[Amount]),
0
)

Max Profit Margin =

(
 MAX('Order Details'[Profit])
)

☐	Monthly Sales
☐	Order Count
☐	Order ID
☐	Σ Profit
☐	Profit-Margin
☐	Σ Quantity
☐	Revenue Per Order
☐	Sales Category

☐	Category
☐	Category Type
☐	Max Profit Margin
☐	Monthly Sales
☐	Order Count

Purpose:

- Find highest margin product groups.

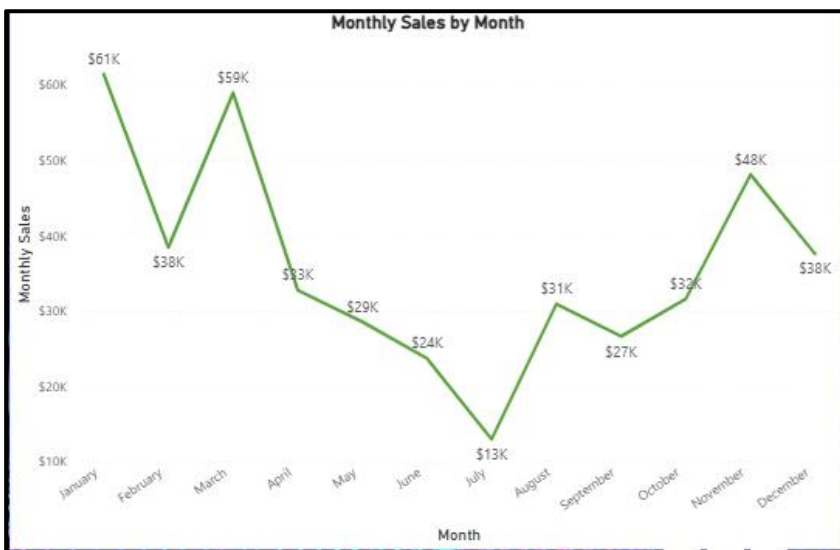
• Monthly Sales Trend:

Show the trend of monthly sales over time using a line chart.

Visual: Line Chart

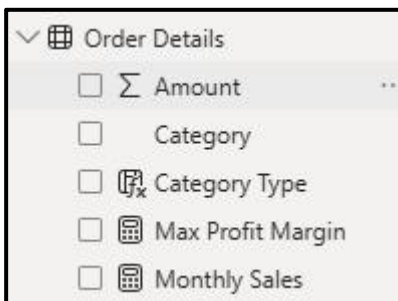
Fields

- X-Axis → Order Date (Month)
- Y-Axis → Total Sales



Measure:

Monthly Sales =
SUM('Order Details'[Amount])



Purpose:

- Observe sales growth and seasonality.

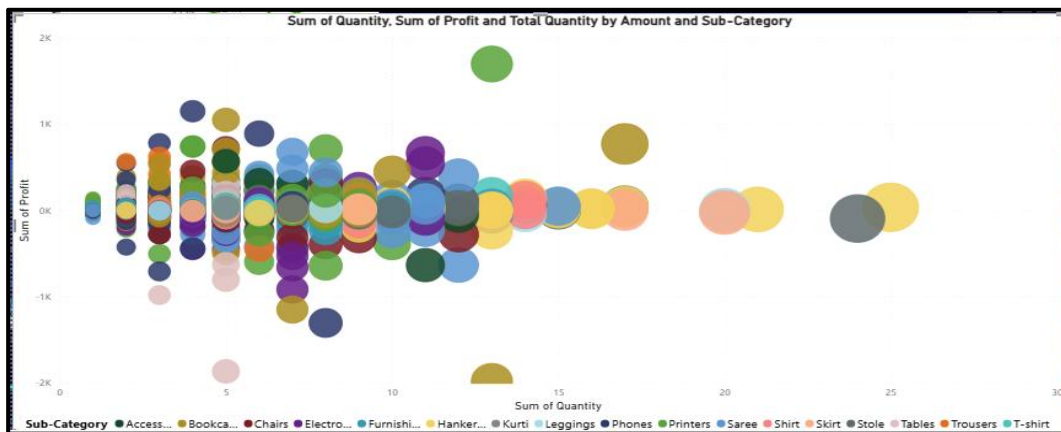
• Comparison of Profit and Quantity by Sub-Category:

Compare the relationship between profit and quantity sold for different sub-categories using a scatter chart.

Visual: Scatter Chart

Fields

- X Axis → Quantity
- Y Axis → Profit
- Details → Sub-Category
- Size → Sales



Measures:

Total Quantity =
`SUM('Order Details'[Quantity])`

Total-Profit =
`SUM('Order Details'[Profit])`

☐ Revenue Per Order ...
 ☐ Sales Category
 ☐ Sub-Category
 ☐ Total Profit
 ☐ Total Quantity
 ☐ Total Sales

Purpose:

- Identify high-profit/high-volume products.

• Comparison of Total Sales Amount and Target:

Create cards to succinctly display the total sales amount alongside the sales target for quick comparison and analysis. Also, create a multi-row card to display the minimum target for each segment.

Visual: Cards



Create:

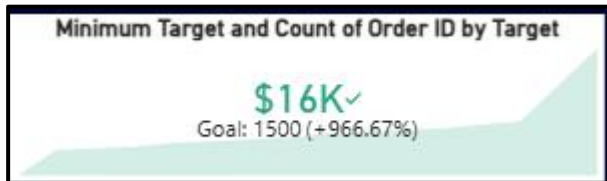
Total Sales Amount =
SUM('Order Details'[Amount])

Sales Target =
SUM('Sales Target'[Target])

Add:

- Card → Total Sales Amount
- Card → Sales Target

● Multi-row Card (Minimum Target by Segment)



Measure:

Minimum Target =
MIN('Sales Target'[Target])

Fields:

- Segment
- Minimum Target

Purpose:

- Quick target comparison.

● Sales Performance Matrix:

Build a matrix view to analyze how actual sales compare to sales targets across different categories and months.

Visual: Matrix

Rows:

- Category

Columns:

- Month

Values:

- Total Sales
- Sales Target

Quantity	Clothing	Electronics	Furniture	Total
1	\$174,000	\$129,000	\$132,900	\$435,900
2	\$174,000	\$129,000	\$132,900	\$435,900
3	\$174,000	\$129,000	\$132,900	\$435,900
4	\$174,000	\$129,000	\$132,900	\$435,900
5	\$174,000	\$129,000	\$132,900	\$435,900
6	\$174,000	\$129,000	\$132,900	\$435,900
7	\$174,000	\$129,000	\$132,900	\$435,900
8	\$174,000	\$129,000	\$132,900	\$435,900
Total	\$174,000	\$129,000	\$132,900	\$435,900

Optional:

Target Achievement % =

DIVIDE(

'Order Details'[Total Sales],

'Sales target'[Total Target],

0

)

Purpose:

- Month-by-month performance tracking.

• Geographic Sales Analysis:

Visualize total sales on a map by city to identify regional sales patterns.

Visual: Map

Fields:

- Location → City
- Size → Total Sales



Measure:

City Sales =
 $\text{SUM}(\text{'Order Details' [Amount]})$

Purpose:

- Identify strongest sales regions.

● Sales Distribution by Sub-Category:

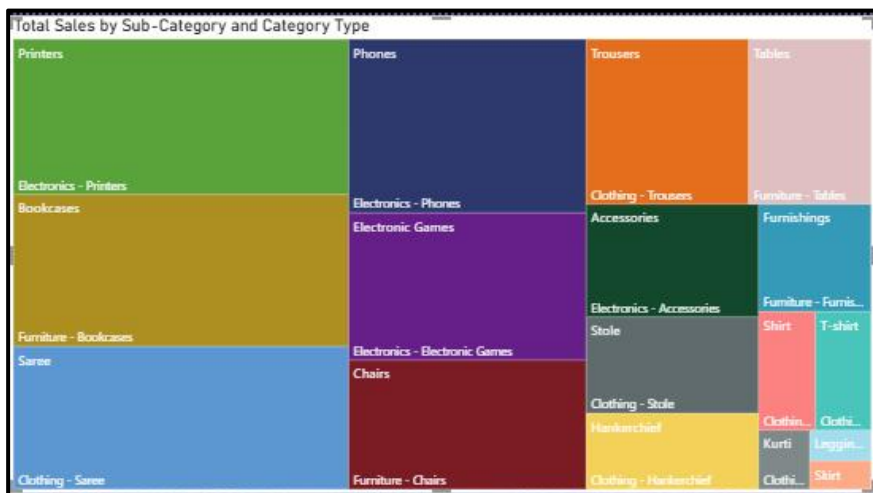
Represent the sales distribution across different

- sub-categories using a tree map.

Visual: Treemap

Fields:

- Group → Sub-Category
- Values → Total Sales



Measure:

Sub Category Sales =
SUM('Order Details'[Amount])

Purpose:

- Understand sales contribution.

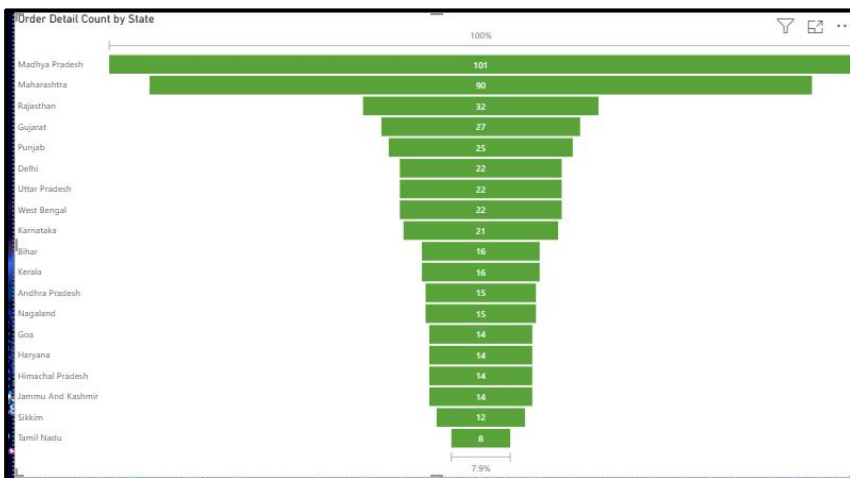
• Order Count Analysis by State:

Create a funnel chart to visualize the distribution of order counts across different states.

Visual: Funnel Chart

Fields:

- Group → State
- Values → Order Count



Measure:

Order Count =
DISTINCTCOUNT('Order Details'[Order ID])

Purpose:

- Compare order volume across states.

● DashBoard

